YouTube Top 5000 Channels Analysis

Project Report

Exploratory analysis of the top 5,000 YouTube channels with Python, pandas and Seaborn

1. Introduction

This project explores a dataset of the 5,000 top-ranked YouTube channels. It asks which channels lead on subscribers, uploads and views, who the biggest music artist channels are, and whether subscribers, views, uploads, rank and grade are related.

2. Objectives

- Clean the data and summarise its numeric columns.
- Rank channels by subscribers, video uploads and total views.
- Identify the leading VEVO (music artist) channels.
- Measure correlations between subscribers, views and uploads.
- Create an average-views-per-video measure and explore it by grade.
- Encode Grade and Rank as numbers and build a correlation matrix.

3. Tools and Technologies

Tool	Purpose
Python	Programming language
pandas / NumPy	Cleaning, sorting, filtering, correlation
Matplotlib / Seaborn	Scatter plots and bar plots
Jupyter / Google Colab	Interactive development environment

4. Dataset

The dataset has **5,000 rows and 6 columns**; no duplicate rows were removed and no column has missing entries. Columns: Rank (text such as '1st' or '1,471st'), Grade, Channel name, Video Uploads, Subscribers and Video views. Video Uploads and Subscribers are stored as text, so they are converted to integers before analysis.

Grade distribution

Grade	A++	A+	A	A-	B+	Blank
Channels	10	41	963	1,024	2,956	6

Summary statistics

Statistic	Video uploads	Subscribers	Video views
Mean	3,855	2,417,216	1.07 billion
Std. dev.	17,076	3,835,852	2.00 billion
Min	0	0	75
25%	140	457,918	186 million
Median	441	1,204,652	482 million
75%	1,500	2,723,182	1.12 billion
Max	422,326	65,995,310	47.5 billion

Every column is strongly right-skewed: the mean is well above the median, so a small number of very large channels pull the averages up.

5. Methodology

- **Load and inspect:** `read_csv`, `drop_duplicates()`, `info()`, `describe()`.
- **Convert:** `pd.to_numeric(errors='coerce')` on Subscribers and Video Uploads, filling failures with 0.
- **Rank:** `sort_values(...).head(10)` by subscribers, uploads and views.
- **Filter:** `str.contains('VEVO')` to find music artist channels.
- **Correlation:** scatter plots and Pearson correlation for three pairs of variables.
- **Feature creation:** Average Views = Video views / Video Uploads.
- **Encode:** Grade mapped to numbers (A++ = 5 down to B+ = 1); Rank cleaned from text (e.g. '1,471st') to digits; full correlation matrix.
- **Visualise:** Seaborn bar plots of subscribers, average views and uploads by grade.

6. Findings

6.1 Top channels

Measure	Leader	Value
Subscribers	PewDiePie (rank 1,471st), then T-Series, Canal KondZilla	65,995,310
Video views	T-Series (then WWE, Ryan ToysReview)	47.5 billion
Video uploads	AP Archive (then YTN NEWS, SBS Drama)	422,326

The leaders differ by measure. The rank column is not simply ordered by subscribers: PewDiePie has the most subscribers yet is ranked 1,471st, so rank clearly depends on other factors. The most prolific uploaders are news, archive and broadcast channels, not the most subscribed ones.

6.2 Music (VEVO) channels

Measure	Top 3
Subscribers	JustinBieberVEVO (34.3M), TaylorSwiftVEVO (28.1M), KatyPerryVEVO (26.7M)
Views	JustinBieberVEVO (17.8B), KatyPerryVEVO (16.2B), TaylorSwiftVEVO (15.5B)

6.3 Correlations

Pair	Correlation	Interpretation
Subscribers vs views	0.79	Strong positive
Subscribers vs uploads	0.02	Negligible
Views vs uploads	0.09	Negligible
Grade (encoded) vs views	0.46	Moderate
Grade (encoded) vs subscribers	0.41	Moderate
Rank vs Grade (encoded)	-0.86	Strong (negative because rank 1 is best)

Channels with more subscribers get far more views, but uploading more videos does not on its own bring more subscribers or views. Average views per video correlates only weakly with subscribers (0.29).

7. Issues and Recommended Improvements

#	Issue	Recommended fix
1	Non-numeric Subscribers and Video Uploads are replaced with 0. Real channels with hidden subscriber counts then appear to have none (the minimum subscribers is 0).	Keep them as missing (NaN) or count and report them; exclude from rankings and means.
2	The Average Views cell drops `avg_views`, which is not a column in the loaded data, and later cells use Average Views from a different DataFrame. A fresh Restart and Run All would raise errors.	Create `Average Views` on the main DataFrame once and remove the `drop('avg_views')` calls.
3	Average Views = views / uploads gives infinity when uploads is 0.	Filter to uploads > 0 or replace infinite values.
4	VEVO filter is called 'music artist channels' but only matches channels with VEVO in the name (misses e.g. Ed Sheeran) and is case-sensitive.	Rename the list 'VEVO channels' or match a broader artist list; use `case=False`.
5	The same numeric-conversion code is repeated in about eight cells.	Convert once after loading.
6	The Average Views vs Grade plot is drawn twice, and `plt.figure(figsize=(100,10))` is called after the plot, creating an empty oversized figure.	Remove the duplicate; set figure size before plotting (e.g. (8, 5)).
7	Grade map keys contain trailing spaces ('A++ ') and the blank grades become NaN. Rank is cleaned by cutting the last two characters.	Strip whitespace first; handle blank grades explicitly; parse Rank with a regular expression.
8	Heavily skewed data is summarised with Pearson correlation only.	Also report Spearman correlation or use log scales on the scatter plots.
9	Data is read from a Colab Drive path.	Use a relative path.

8. Conclusion

The analysis shows that top YouTube channels are very unevenly sized: a few giants such as T-Series, PewDiePie and WWE dominate subscribers and views while the median channel has about 1.2 million subscribers. Subscribers and views move together strongly (0.79), whereas the number of uploads has almost no relationship with either. Grade tracks rank closely and rises moderately with subscribers and views. Handling hidden subscriber counts properly and fixing the Average Views steps would make the notebook more accurate and able to run from start to finish.

9. Future Work

- Plot distributions on a log scale and add a correlation heatmap.
- Categorise channels (music, news, kids, entertainment, sports) and compare them.
- Study subscribers per video and views per subscriber as engagement measures.
- Examine how grade relates to subscribers and views (grade thresholds).
- Look at channel-name patterns (such as 'Topic' auto-generated channels, VEVO, 'Official').
- Build a model to predict views from subscribers and uploads.